

Asunción Jiménez-Cordero

Curriculum Vitae

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Personal Information

Name	JIMÉNEZ CORDERO, M. Asunción
Position	Associate Professor (Profesora Titular) at the Department of Mathematical Analysis, Statistics and Operations Research and Applied Mathematics. Area of research: Statistics and Operations Research. University of Málaga.
Address	Ada Byron Research Building, Calle Arquitecto Francisco Peñalosa, 18, 29010, Málaga, Spain.
Telephone	+34 951 95 29 25 / +34 951 95 34 21 / +34 952 13 18 96
E-mail	asuncionjc@uma.es
Nationality	Spanish
Maternity Leaves	Sep 2026 – Oct 2026 and Feb 2026 – Jun 2026
ResearchGate	`https://www.researchgate.net/profile/Asuncion_Jimenez-Cordero`
Google Scholar	`https://scholar.google.es/citations?user=JegcEYwAAAAJ&hl=es&oi=ao`
Github	`https://github.com/asuncionjc`
ORCID	`https://orcid.org/0000-0001-6551-692X`

Education

Feb 2019 **Ph.D. in Mathematics**, *University of Seville*, International Doctorate Mention.
Title: Classification and Regression with Functional Data. A Mathematical Optimization Approach.
Calification: Highest distinction (Suma Cum laude).
Supervisors: Dr. Rafael Blanquero and Dr. Emilio Carrizosa.
Outstanding Doctoral Thesis Award.

Oct 2013 – Jul 2014 **Master degree in Advanced Mathematics**, *University of Seville*, *GPA – 9.25*.

Oct 2009 – Jul 2013 **Bachelor in Mathematics (240 ECTS Credits)**, *University of Seville*, *GPA– 8.8*.
Degree Thesis with honours.
Best Student in Bachelor in Mathematics

Fellowships, Contracts and Positions

Since Jan 2026 **Associate Professor (Profesora Titular)**, *Department of Mathematical Analysis, Statistics and Operations Research and Applied Mathematics*, Area of research: Statistics and Operations Research, University of Málaga.

Oct 2024 - Jan 2026 **Senior Lecturer (Profesora Permanente Laboral)**, *Department of Mathematical Analysis, Statistics and Operations Research and Applied Mathematics*, Area of research: Statistics and Operations Research, University of Málaga.

Jan 2022 - Oct 2024 **Lecturer (Assistant Professor)**, *Department of Mathematical Analysis, Statistics and Operations Research and Applied Mathematics*, Area of research: Statistics and Operations Research, University of Málaga.

Dec 2021 **Postdoctoral position Andalusia Regional Government (Junta de Andalucía)**.
Renounce to take the lecturer position at the University of Málaga.

Apr 2019 - Jan 2022 **Post-doctoral research contract in the ERC project "Advanced Analytics to Empower the Small Flexible Consumers of Electricity (FlexAnalytics) (8.06.UE/33.8039)"**, *OASYS group (University of Málaga)*.

Jul 2017 **Grant devoted to the Research Stays in Foreign Universities (Ayudas para Estancias Breves en Otras Universidades)**, *Research Office of the University of Seville (Vicerrectorado de Investigación de la Universidad de Sevilla)*.

Mar 2017 **Grant of Project COST Action TD 1207**, *Optimization Training School Novi Sad Faculty of Technical Sciences*.

Jun 2016 **Grant devoted to the International Movements of the Research Staff (Ayudas para Movilidad Internacional del Personal dedicado a la Investigación)**, *Research Office of the University of Seville (Vicerrectorado de Investigación de la Universidad de Sevilla)*.

Jan 2016 – Apr 2016 **Participation as researcher in the Project contract "Desarrollo de un algoritmo para la optimización de rutas de reparto"**, *FIUS (University of Seville)*.

Sep 2015- Dec 2018 **Spanish Ministry of Education PhD Fellowship (Formación del Profesorado Universitario - FPU)** , *Mathematical Research Institute of the University of Seville (IMUS) and Department of Statistics and Operations Research (University of Seville)*.

Mar 2015 **Grant of Project COST Action TD 1207**, *Optimization Training School Tilburg University*.

Jan 2015 – Aug 2015 **Cámara Foundation PhD Fellowship**, *Mathematical Research Institute of the University of Seville (IMUS) and Department of Statistics and Operations Research (University of Seville)*.
Renounce to take Spanish Ministry of Education, Science and Sports Grant

Jan 2015 – Feb 2015 **Grant of Project COST Action TD 1207**, *Stay in Budapest University of Technology and Economics (Hungary) with the Prof. Boglárka G. Tóth*.

Dec 2014 – Dec 2015 **Participation as researcher in the project "Identificación de Modelos en Procesos con Reacción Química. Fase II"**, *(REPSOL-FIUS-IMUS)* .

Oct 2014 – Dec 2014 **Research grant for PhD students (Beca de Iniciación a la Investigación - Modalidad B)**, *Mathematical Research Institute of the University of Seville (IMUS)*.

Dec 2013 – Aug 2014 **Project contract "Identificación de Modelos en Procesos con Reacción Química"**, *REPSOL - FIUS - IMUS*.

Oct 2013 – Nov 2013 **Research grant for graduate students (Beca de Iniciación a la Investigación - Modalidad A)**, *Mathematical Research Institute of the University of Seville (IMUS)*.

Oct 2012 – Jun 2013 **Collaborator (alumna interna)**, *Department of Statistics and Operations Research*, University of Seville.

Oct 2012 – Jun 2013 **Research collaboration grant**, *Spanish Ministry of Education, Science and Sports*.

Oct 2011 – Jun 2012 **Collaborator (alumna interna)**, *Department of Mathematical Analysis*, University of Seville.

Publications

[1] **An enhanced heuristic framework for solving the Rank Pricing Problem**, *Asunción Jiménez-Cordero, Salvador Pineda, Juan Miguel Morales*, Expert Systems with Applications, 2025, 276, 127122, doi: 10.1016/j.eswa.2025.127122.
Impact Factor: 9.4. Position: 6 out 109. Rank: D1. Area: Operations Research and Management Science. Source: JCR 2025.

[2] **Learning-Assisted Optimization for Transmission Switching**, *Salvador Pineda, Juan Miguel Morales, Asunción Jiménez-Cordero*, TOP - Transactions on Operations Research, 2024, 32, 489-516, doi: 10.1007/s11750-024-00672-0.
Impact Factor: 1.4. Position: 82 out 106. Rank: Q4. Area: Operations Research and Management Science. Source: JCR 2024.

[3] **Cost-driven Screening of Network Constraints for the Unit Commitment Problem**, *Álvaro Porras-Cabrera, Salvador Pineda, Juan Miguel Morales, Asunción Jiménez-Cordero*, IEEE Transactions on Power Systems, 2023, 38(1), 42-51, doi: 10.1109/TPWRS.2022.3160016.
Impact Factor: 6.5. Position: 40 out of 352. Rank: Q1. Area: Engineering, Electrical and Electronic. Source: JCR 2023.

[4] **Warm-starting constraint generation for mixed-integer optimization: A Machine Learning approach**, *Asunción Jiménez-Cordero, Juan Miguel Morales, Salvador Pineda*, Knowledge-Based Systems, 2022, 253, 109570, doi: 10.1016/j.knosys.2022.109570.

Impact Factor: 8.8. Position: 19 out 145. Rank: Q1. Area: Computer Science, Artificial Intelligence. Source: JCR 2022.

[5] **A novel embedded min-max approach for feature selection in nonlinear Support Vector Machine classification**, *Asunción Jiménez-Cordero, Juan Miguel Morales, Salvador Pineda*, European Journal of Operational Research, 2021, 293(1), 24-35, doi: 10.1016/j.ejor.2020.12.009.

Impact Factor: 5.334. Position: 15 out of 84. Rank: Q1. Area: Operations Research and Management Science. Source: JCR 2021.

[6] **Automatic Feature Scaling and Selection for Support Vector Machine Classification with Functional Data**, *Asunción Jiménez-Cordero, Sebastián Maldonado*, Applied Intelligence, 2021, 51, 161-184, doi: 10.1007/s10489-020-01765-6.

Impact Factor: 5.086. Position: 35 out of 139. Rank: Q2 (T1). Area: Computer Science, Artificial Intelligence. Source: JCR 2020.

[7] **Selection of Time Instants and Intervals with Support Vector Regression for Multivariate Functional Data**, *Rafael Blanquero, Emilio Carrizosa, Asunción Jiménez-Cordero, Belén Martín-Barragán*, Computers and Operations Research, 2020, 123, 105050, doi: 10.1016/j.cor.2020.105050.

Impact Factor: 4.008. Position: 25 out of 84. Rank: Q2 (T1). Area: Operations Research and Management Science. Source: JCR 2020.

[8] **Data-Driven Screening of Network Constraints for Unit Commitment**, *Salvador Pineda, Juan Miguel Morales, Asunción Jiménez-Cordero*, IEEE Transactions on Power Systems, 2020, 35(5), 3695 - 3705, doi: 10.1109/TPWRS.2020.2980212.

Impact Factor: 6.663. Position: 26 out of 273. Rank: Q1 (D1). Area: Engineering, Electrical and Electronic. Source: JCR 2020.

[9] **Variable Selection in Classification for Multivariate Functional Data**, *Rafael Blanquero, Emilio Carrizosa, Asunción Jiménez-Cordero, Belén Martín-Barragán*, Information Sciences, 2019, 481, 445-462, doi: 10.1016/j.ins.2018.12.060.

Impact Factor: 5.91. Position: 9 out of 156. Rank: Q1 (D1). Area: Computer Science, Information Systems. Source: JCR 2019.

[10] **Functional-bandwidth Kernel for SVM with Functional Data: an Alternating Optimization Algorithm**, *Rafael Blanquero, Emilio Carrizosa, Asunción Jiménez-Cordero, Belén Martín-Barragán*, European Journal of Operational Research, 2019, 275(1), 195-207, doi: 10.1016/j.ejor.2018.11.024.

Impact Factor: 4.213. Position: 14 out of 83. Rank: Q1. Area: Operations Research and Management Science. Source: JCR 2019.

[11] **A Global Optimization Method for Model Selection in Chemical Reactions Networks**, *Rafael Blanquero, Emilio Carrizosa, Asunción Jiménez-Cordero, José Francisco Rodríguez*, Computers & Chemical Engineering, 93, 2016, 52-62, doi: 10.1016/j.compchemeng.2016.05.016.

Impact Factor: 3.024. Position: 20 out of 105. Rank: Q1. Area: Computer Science and Interdisciplinary Applications. Source: JCR 2016.

[12] **On Extreme Concentrations in Chemical Reaction Networks with Incomplete Measurements**, *Rafael Blanquero, Emilio Carrizosa, Oana Chis, Noemí Esteban, Asunción Jiménez-Cordero, José Francisco Rodríguez, M. Remedios Sillero-Denamiel*, Industrial & Engineering Chemistry Research, 2016, 55 (44), 11417-11430, doi: 10.1021/acs.iecr.6b00714.
Impact Factor: 2.843. Position: 34 out of 135. Rank: Q2 (T1). Area: Engineering, Chemical. Source: JCR 2016.

Working Papers

[1] **A Class of Rényi-based Concentration Metrics for Kolmogorov-Arnold Geometry in Multilayer Perceptrons**, *Juan Carlos Fernández, Asunción Jiménez-Cordero, José Franscisco Paris*, Submitted, 2026.
Available at `https://www.researchgate.net/publication/416253314_A_Class_of_Renyi-based_Concentration_Metrics_for_Kolmogorov-Arnold_Geometry_in_Multilayer_Perceptrons`

Teaching Publications

[1] **Gender perspectives in STEM education. An exploratory study with pre-service teachers**, *Cristina García-Ruiz, Verónica Torres-Blanco, Desirée García-Durán, Carolina Martín-Gámez, Agustina Torres-Prioris, Asunción Jiménez-Cordero*, International Journal of Science Education, 2026, https://doi.org/10.1080/09500693.2026.2727689.
Impact Factor: 2.4. Position: 191 out 760. Rank: Q2 (T1). Area: Eduaction and Educational Research. Source: JCR 2025.

Awards, Recognitions and Qualifications

Sep 2024 **Qualification Associate Professor (Acreditación Profesor Titular de Universidad)**, *National Agency of Quality Evaluation and Acreditation (Agencia Nacional de Evaluación de la Calidad y Acreditación - ANECA*.

Nov 2021 **YoungWomen4OR Award**, *EURO - The Association of European Operational Research Societies*, WISDOM - Women in Science Doing Operational Research and Management.

Jul 2021 **Outstanding Doctoral Thesis Award**, *PhD in Mathematics*, University of Seville.

Dec 2020 **Qualification Senior Lecturer (Acreditación Contratado Doctor)**, *National Agency of Quality Evaluation and Acreditation (Agencia Nacional de Evaluación de la Calidad y Acreditación - ANECA)*.

Oct 2019 **Qualification Lecturer (Acreditación Ayudante Doctor)**, *National Agency of Quality Evaluation and Acreditation (Agencia Nacional de Evaluación de la Calidad y Acreditación - ANECA)*.

Mar 2018 **Elsevier Reviewer Recognition**, *Association of European Operational Research Societies (EURO) and The International Federation of Operational Research Societies (IFORS)*.

Oct 2014 **Extraordinary End-of-bachelor Award**, *University of Seville*.

Peer-Review

Jun 2025 **Reviewer International Transactions in Operations Research (ITOR)**.
Oct 2024 **Reviewer Computers and Operations Research (CAOR)**.
Oct 2024 **Reviewer Applied Energy (APEN)**.
Feb 2024 **Reviewer Information Sciences (IS)**.
Sep 2022 **Reviewer European Journal of Operational Research (EJOR)**.
Jul 2022 **Reviewer Computers and Operations Research (CAOR)**.
Jul 2022 **Reviewer European Journal of Operational Research (EJOR)**.
Jul 2021 **Reviewer European Journal of Operational Research (EJOR)**.
Jun 2021 **Reviewer Artificial Intelligence Review**.
Apr 2021 **Reviewer European Journal of Operational Research (EJOR)**.
Jul 2020 **Reviewer IEEE Transactions on Power Systems**.
Jul 2020 **Reviewer Advances in Data Analysis and Classification**.
Apr 2020 **Reviewer IISE Transactions**.
May 2019 **Reviewer European Journal of Operational Research (EJOR)**, *(twice)*.
Mar 2018 **Reviewer European Journal of Operational Research (EJOR)**.

Stays

Jun 2018 **Copenhagen Business School**, *Copenhagen, Denmark*, Prof. Dolores Romero Morales, 1 week.

Sep 2017 **University of Chile (Universidad de Chile)**, *Santiago, Chile*, Prof. Richard Weber, 3 months.

Oct 2016 **University of Edinburgh. Business School**, *Edinburgh, Scotland*, Prof. Belén Martín-Barragán, 1 week.

Jan 2015 **Budapest University of Technology and Economics**, *Budapest, Hungary*, Prof. Boglárka G. Tóth, 1 month.

Contribution to Conferences

Jun 2025 **Solving the Rank Pricing Problem using a heuristic approach**, *XLI Congreso Nacional de Estadística e Investigación Operativa (SEIO 2025)*, *Asunción Jiménez-Cordero*, Salvador Pineda and Juan Miguel Morales, Seville, Spain.
Invited talk

Sep 2024 **On solving the Rank Pricing Problem using a VNS-based approach**, *Mathematical Optimization and Statistics for Explainable and Fair Machine Learning (Edition 2024)*, *Asunción Jiménez-Cordero*, Salvador Pineda and Juan Miguel Morales, Seville, Spain.
Invited talk

Jul 2024 **A data-based approach for solving the Rank Pricing Problem**, *The 33rd European Conference on Operational Research (EURO 2024)*, *Asunción Jiménez-Cordero*, Salvador Pineda and Juan Miguel Morales, Copenhagen, Denmark.
Invited talk

Jul 2024 **Learning-Assisted Optimization for Transmission Switching**, *The 33rd European Conference on Operational Research (EURO 2024)*, *Salvador Pineda*, Juan Miguel Morales and Asunción Jiménez-Cordero, Copenhagen, Denmark.
Invited talk

Nov 2023 **Machine-Learning-aided Optimal Transmission Switching**, *XL National Conference of Statistics and Operations Research (XL Congreso Nacional de Estadística e Investigación Operativa, SEIO 2023)*, Salvador Pineda, Juan Miguel Morales and *Asunción Jiménez-Cordero*, Elche, Spain.
Invited talk

Jun 2023 **A novel machine learning approach for solving optimal transmission switching**, *3rd EUROYoung Workshop*, Salvador Pineda, Juan Miguel Morales and *Asunción Jiménez-Cordero*, Cergy, Paris, France.
Invited talk

Sep 2022 **Solving mixed-integer programs with warm-starting constraint generation methods via machine learning tools**, *3rd Spanish Young Statisticians and Operational Researchers Meeting (SYSORM)*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Elche, Spain.
Invited talk

Jul 2022 **Feature Selection in (Functional) SVM via Bilevel Optimization**, *The 32nd European Conference on Operational Research (EURO 2022)*, *Asunción Jiménez-Cordero*, Espoo, Finland.
Invited talk

Jul 2022 **A novel machine-learning-aided approach for warm-starting constraint generation methods in MILPs**, *The 32nd European Conference on Operational Research (EURO 2022)*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Espoo, Finland.
Invited talk

Jul 2022 **Data-driven screening of network constraints for Unit Commentment**, *2022 IEEE Power & Energy Society General Meeting (GM)* , *Salvador Pineda*, Juan Miguel Morales, Asunción Jiménez-Cordero, Denver, USA.
Invited talk

Jun 2022 **On warm-starting constraint generation methods via machine learning tools for solving mixed-integer programs**, *XXXIX National Conference of Statistics and Operations Research (XXXIX Congreso Nacional de Estadística e Investigación Operativa, SEIO 2022)*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Granada, Spain.
Invited talk

May 2022 **On selecting features in (functional) SVM with bilevel optimization tools**, *WISDOM Webinar on Bilevel Optimization*, *Asunción Jiménez-Cordero*, Online.
Invited talk

Dec 2021 **Classifying (functional) data from a mathematical optimization perspective. Application to real-life problems**, *Outstanding Doctoral Thesis Award Seminar (IMUS)*, *Asunción Jiménez-Cordero*, Seville, Spain.
Invited talk

Nov 2021 **An offline-online strategy to improve MILP performance via Machine Learning tools**, *Workshop "New Bridges between Mathematics and Data Science"*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Valladolid, Spain.
Invited talk

Oct 2021 **Machine learning-aided optimization applied to mixed integer linear programming**, *Seminar "Operations Research"*, *Antonio Elías-Fernández, Asunción Jiménez-Cordero*, Málaga, Spain.
Talk

Jul 2021 **Improving the performance of machine learning in the solution of MILPs**, *The 31st European Conference on Operational Research (EURO 2021)*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Online.
Invited talk

Jul 2021 **Cost-aware constraint screening for the unit commitment problem**, *The 31st European Conference on Operational Research (EURO 2021)*, *Álvaro Porras Cabrera*, Salvador Pineda, Juan Miguel Morales, Asunción Jiménez-Cordero, Online.
Invited talk

Jul 2021 **Data-driven screening of network constraints for Unit Commitment**, *IEEE PowerTech Conference*, *Salvador Pineda*, Juan Miguel Morales, Asunción Jiménez-Cordero, Online.
Talk

Jun 2021 **A novel embedded min-max approach for feature selection in nonlinear Support Vector Machine classification**, *SEIO 2021*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Online.
Talk

Feb 2021 **Feature selection in nonlinear SVM using a novel minmax approach**, *Young Online Seminar Series Machine Learning NeEDS Mathematical Optimization*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Online.
Talk

Nov 2020 **Interpretable Learning in Power Systems Operations**, *Informs 2020 Annual Meeting*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Online.
Invited talk

Nov 2020 **Data-driven Screening of Network Constraints for Unit Commitment**, *Informs 2020 Annual Meeting*, *Salvador Pineda*, Juan Miguel Morales, Asunción Jiménez-Cordero, Online.
Talk

Oct 2020 **A novel min-max approach to select features in nonlinear SVM classification**, *Autumn School on Bilevel Optimization 2020*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Online.
Talk

Jan 2020 **A min-max approach to feature selection for nonlinear SVM classification**, *V Conference of Young Researchers RSME 2020*, *Asunción Jiménez-Cordero*, Juan Miguel Morales, Salvador Pineda, Castellón, Spain.
Invited talk

Sept 2018 **On Variable Selection in Supervised Learning for Functional Data**, *Workshop on Data Science*, Rafael Blanquero, Emilio Carrizosa, *Asunción Jiménez-Cordero*, Sebastián Maldonado, Belén Martín-Barragán, Seville, Spain.
Talk

Jul 2018 **Feature Selection with SVM and SVR for Functional Data**, *16th Workshop on Advances in Continuous Optimization (EUROPT 2018)*, Rafael Blanquero, Emilio Carrizosa, *Asunción Jiménez-Cordero*, Sebastián Maldonado, Belén Martín-Barragán, Almería, Spain.
Talk

Jul 2018 **Some improvements of SVM and SVR for Functional Data**, *29th European Conference on Operational Research (EURO 2018)*, Rafael Blanquero, Emilio Carrizosa, *Asunción Jiménez-Cordero*, Belén Martín-Barragán, Valencia, Spain.
Invited talk

Jun 2018 **Mejoras en SVM y SVR con datos funcionales**, *XXXVII Congreso Nacional de Estadística e Investigación Operativa*, Rafael Blanquero, Emilio Carrizosa, *Asunción Jiménez-Cordero*, Belén Martín-Barragán, Oviedo, Spain.
Talk

Nov 2017 **Improving SVM-classification with Functional Data**, *XII Chilean Congress of Operational Research (XII Congreso Chileno de Investigación Operativa - Optima 2017)*, Rafael Blanquero, Emilio Carrizosa, *Asunción Jiménez-Cordero*, Belén Martín-Barragán, Viña del Mar, Chile.
Talk

Sep 2017 **On SVM-classification with Functional Data**, *XVI Spanish Biometric Conference (CEB 2017)*, *Rafael Blanquero*, Emilio Carrizosa, Asunción Jiménez-Cordero, Belén Martín-Barragán, Seville, Spain.
Talk

Jul 2017 **On Support Vector Machines for Functional Data**, *15th EUROPT Workshop on Advances in Continuous Optimization*, Rafael Blanquero, *Emilio Carrizosa*, Asunción Jiménez-Cordero, Belén Martín-Barragán, Montreal, Canada.
Talk

Jul 2017 **Some improvements in the SVM-Classification with Functional Data**, *Summer School: Optimization, Big Data and Applications*, Rafael Blanquero, Emilio Carrizosa, *Asunción Jiménez-Cordero*, Belén Martín-Barragán, Veroli, Italy.
Talk

Jan 2017 **Model Selection and Missing Data in Processes Governed by ODEs**, *Workshop Recent Advances in PDEs: Analysis, Numerics and Control*, Rafael Blanquero, Emilio Carrizosa, Oana Chís, Noemí Esteban, *Asunción Jiménez Cordero*, José Francisco Rodríguez, M. Remedios Sillero Denamiel, Seville, Spain.
Poster

Jan 2017 **Design of an APP-based System for VRP with Time Windows**, *Avances en problemas de logística y problemas de rutas. Modelos con uno o varios objetivos*, *Rafael Blanquero*, Emilio Carrizosa, M. Asunción Jiménez Cordero, Seville, Spain.
Talk

Sep 2016 **Model Selection and Missing Data in Chemical Reactions Networks. A Global Optimization Approach.**, *XXXVI Congreso Nacional de Estadística e Investigación Operativa (SEIO)*, Rafael Blanquero, Emilio Carrizosa, Oana Chis, Noemí Esteban, *M. Asunción Jiménez Cordero*, José Francisco Rodríguez, M. Remedios Sillero Denamiel, Toledo, Spain.
Talk

July 2016 **A MINLP to Parameters Inference in Chemical Reactions Networks**, *The 28th European Conference on Operational Research (EURO)*, Rafael Blanquero, Emilio Carrizosa, José Luis Ferrín, *M. Asunción Jiménez Cordero*, José Francisco Rodríguez, Jerónimo Rodríguez, Boglárka G. Tóth, Poznan, Poland.
Talk

July 2016 **Global Optimization Tools to Discriminate Models in Chemical Reaction Networks**, *The 14th Workshop on Advances in Continuous Optimization (EUROPT)*, Rafael Blanquero, Emilio Carrizosa, *M. Asunción Jiménez Cordero*, José Francisco Rodríguez, Warsaw, Poland.
Talk

June 2016 **Automatic Identification of Kinetic Models in Industrial Reaction Systems**, *The 19th European Conference on Mathematics for Industry (ECMI 2016)*, Gabriel Álvarez, Marta Benítez, Alfredo Bermúdez, Rafael Blanquero, Emilio Carrizosa, Oana Chis, Oscar Crego, Manuel Cremades, Noemí Esteban, José Luis Ferrín, M. Asunción Jiménez Cordero, Francisco Pena, Diego Rodríguez, Jerónimo Rodríguez, *José Francisco Rodríguez*, M. Remedios Sillero Denamiel, Santiago de Compostela, Spain.
Talk

June 2016 **Parameters Inference in Chemical Reactions Networks. Model Selection and Missing Data**, *The 19th European Conference on Mathematics for Industry (ECMI 2016)*, Rafael Blanquero, Emilio Carrizosa, Oana Chis, Noemí Esteban, *M. Asunción Jiménez Cordero*, José Francisco Rodríguez, M. Remedios Sillero Denamiel, Santiago de Compostela, Spain.
Talk

Nov 2015 **A Dispersion Model in ODEs**, *VI International Workshop on Locational Analysis and Related Problems*, Rafael Blanquero, Emilio Carrizosa, *M. Asunción Jiménez Cordero*, Boglárka G. Tóth, Barcelona, Spain.
Talk

Sep 2015 **Global Optimization for Model Discrimination with ODEs**, *CORAL (Combinatorial Optimization, Routing and Location)*, Rafael Blanquero, Emilio Carrizosa, *M. Asunción Jiménez Cordero*, Boglárka G. Tóth, Salamanca, Spain.
Talk

Jan 2015 **Global Optimization Methods for Inferring Stoichiometric Matrices in Chemical Reaction Networks**, *Institute of Informatics at the University of Szeged*, Rafael Blanquero, Emilio Carrizosa, *M. Asunción Jiménez Cordero*, Boglárka G. Tóth, Szeged, Hungary.
Talk

Nov 2014 **An Optimization Problem in Chemical Reaction Networks**, *PhD Seminar of Mathematical Research Institute of the University of Seville (IMUS)*, Rafael Blanquero, Emilio Carrizosa, *M. Asunción Jiménez Cordero*, Boglárka G. Tóth, Seville, Spain.
Talk

Nov 2014 **MINLP Problems in Inference of Chemical Reaction Networks**, *Informs Annual Meeting 2014*, Rafael Blanquero, *Emilio Carrizosa*, M. Asunción Jiménez Cordero, Boglárka G. Tóth, San Francisco, USA.
Talk

Sep 2014 **Global Optimization in Inference of Chemical Reaction Networks**, *XII International Workshop on Global Optimization 2014*, Rafael Blanquero, Emilio Carrizosa, *M. Asunción Jiménez Cordero*, Boglárka G. Tóth, Málaga, Spain.
Talk

Jul 2014 **Computing Stoichiometric Matrices in Chemical Reactions. An MINLP Problem**, *20th Conference of the International Federation of Operational Research Societies*, Rafael Blanquero, *Emilio Carrizosa*, M. Asunción Jiménez Cordero, Boglárka G. Tóth, Barcelona, Spain.
Talk

May 2014 **Inferencia en Reacciones Químicas Mediante Métodos de Optimización Entera**, *VI Congreso Internacional en Formación y Modelización en Ciencias Básicas*, Rafael Blanquero, *Emilio Carrizosa*, M. Asunción Jiménez Cordero, Boglárka G. Tóth, Medellín, Colombia.
Talk

Contribution to Teaching Conferences

Aug 2026 **Design and validation of an instrument for analysing gender perspectives in STEM teaching practices: The WOUMAP questionnaire**, *ESERA 2025: Transitions in Science Education: Sustainability and Digital Advances*, *Cristina García-Ruiz*, Carolina Martín-Gámez, Asunción Jiménez-Cordero, Agustina Torres-Prioris, Verónica Torres-Blanco and Desirée García-Durán, Copenhagen, Denmark.
Talk

Sept 2026 **Percepción de profesorado en formación inicial sobre aspectos de su identidad docente STEM con mperspectiva de género**, *ESERA 2025: Transitions in Science Education: Sustainability and Digital Advances*, *Carolina Martín-Gámez*, Desirée García-Durán, Agustina Torres-Prioris, Asunción Jiménez-Cordero, Cristina García-Ruiz and Verónica Torres-Blanco , Bilbao, Spain.
Talk

Research Courses

Nov 2022 **Workshop on Quasi-Newton Methods**, *Edinburgh, United Kingdom*, (Online attendance).

Sep 2022 **The FlexAnalytics Symposium on Data-driven Decision Making via Mathematical Optimization and Statistics**, *Málaga, Spain*.

Jun 2021 **Spring School on MINLPs and Bilevel Problems 2021**, *Online*, Research group: Polyhedra and Combinatorial Optimization, Martin Schmidt and Yasmine Beck.

Jun 2021 **High Performance Computing for IAMAT cluster**, *Online*, 16 hours.

Dec 2020 **Georgia Tech Virtual Workshop on Energy Systems and Optimization**, *Online*.

Oct 2020 **Autumn School on Bilevel Optimization 2020**, *Online*.

Jun 2019 **Workshop Design and Simulation of Deep Neural Networks using Keras and Tensorflow in Python**, *University of Málaga*, Catalin Stoean and Ruxandra Stoean, Málaga, Spain.

Sep 2018 **An Introduction to Arc Routing**, *Mathematical Research Institute of the University of Seville (IMUS)*, Gilbert Laporte, Seville, Spain.

May 2018 **Equilibrium Analysis for Hierarchical Stochastic Models and Applications**, *Mathematical Research Institute of the University of Seville (IMUS)*, Lina Mallozzi, Seville, Spain.

Mar 2018 **Version Control Systems for Mathematical Researchers**, *Mathematical Research Institute of the University of Seville (IMUS)*, Pedro V. Gómez Sánchez, Seville, Spain.

Jul 2017 **Summer School: Optimization, Big Data and Applications**, *University of Florence*, Veroli, Italy.

Apr 2017 **The complexity of simple modes - a study of worst and typical hard cases of the StQP**, *Mathematical Research Institute of the University of Seville (IMUS)*, Immanuel Bomze, Seville, Spain.

Dec 2016 **Doc Course Optimization**, *Mathematical Research Institute of the University of Seville (IMUS)*, Victor Blanco Izquierdo, Ángel Corberán Salvador, Alfredo Marín Pérez, Juan Francisco Monge Ivars, Miguel A. Pozo and Antonio Rodríguez Chía, Seville, Spain.

Nov 2016 **Análisis de Big Data: retos y oportunidades**, *Mathematical Research Institute of the University of Seville (IMUS)*, Daniel Peña Sanchéz de Rivera, Seville, Spain.

May 2016 **Máquinas de Vectores Soporte**, *Mathematical Research Institute of the University of Seville (IMUS)*, Luis González Abril, Seville, Spain.

Dec 2015 **Workshop on Mathematical Technology transfer: CTA-IMUS-Math-in**, *Seville, Spain*.

Dec 2015 **II Pre-Doc Course on Operations Research 2015 (2nd part)** , *Mathematical Research Institute of the University of Seville (IMUS)*, Alberto Muñoz García and Victor Blanco Izquierdo, Seville, Spain.

Oct 2015 **Técnicas Predictivas en Estadística: Aplicaciones en Kaggle**, *Mathematical Research Institute of the University of Seville (IMUS)*, José Antonio Guerrero Durán, Seville, Spain.

Sep 2015 **Uncertainty in Discrete Location**, *Mathematical Research Institute of the University of Seville (IMUS)*, Elena Fernández, Seville, Spain.

Jun 2015 **II Pre-Doc Course on Operations Research 2015** , *Mathematical Research Institute of the University of Seville (IMUS)*, Alfredo Marín, Dolores Romero Morales and Pepa Ramírez Cobo, Seville, Spain.

May 2015 **IX Reunión del Grupo Español de Decisión Multicriterio**, *Grupo Español de Decisión Multicriterio*, Seville, Spain.

Mar 2015 **MINO/COST Spring School Course on Mixed Integer Nonlinear Optimization and Applications**, *Tilburg University*, Tilburg, Netherlands.

Nov 2014 **Cooperative Game Theoretic Models: Applications to railway systems and collection and disposal of urban solid waste**, *Mathematical Research Institute of the University of Seville (IMUS)*, Domenico Fragnelli, Seville, Spain.

Oct - Nov 2014 **Doc Course Optimization**, *Mathematical Research Institute of the University of Seville (IMUS)*, Martine Labbé, Elena Fernández, Francisco Saldanha, and Jörg Kalcsics , Seville, Spain.

Oct 2014 **International Workshop on Locational Analysis and Related Problems 2014**, *Spanish Location Network and Location Group GELOCA (SEIO) and Mathematical Research Institute of the University of Seville (IMUS)*, Seville, Spain.

Oct 2014 **Network Design in Phylogenetics**, *Mathematical Research Institute of the University of Seville (IMUS)*, Martine Labbé, Seville, Spain.

Jul 2014 **Algebraic and Geometric Techniques in Optimization**, *Mathematical Research Institute of the University of Seville (IMUS)*, Jesús Antonio de Loera, Seville, Spain.

Dec 2013 **Pre-Doc Course on Operations Research**, *Mathematical Research Institute of the University of Seville (IMUS)*, Dolores Romero Morales and Boglárka G. Tóth, Seville, Spain.

Oct 2013 **Metaheurísticas**, *Mathematical Research Institute of the University of Seville (IMUS)*, José Andrés Moreno Pérez, Seville, Spain.

Participation in Research Projects

As Principal Investigator

May 2021 - May 2023 *MLO-MILP: "Machine Learning-aided Optimization applied to Mixed Integer Linear Programming"*. Ref. B1-2020-15, 2021-2023, 4000 €. University of Málaga.

As Member of the Research Team

Since Jan 2026 *"Blitz Optimization for Power System Operation" (BlitzPower)* . Junta de Andalucía. Principal Investigators: Juan Miguel Morales González and Salvador Pineda Morente.

Since Jan 2025 *"Mathematical Methods for the Optimization of Decision-making Processes based on Multi-source Data" (MSOpt)* , (1/01/2025 - 31/12/2028). Spanish Ministry of Innovation and Science. Principal Investigators: Juan Miguel Morales González and Salvador Pineda Morente.

Sep 2021 - Feb 2025 *"Data-driven Optimization Under a Dynamic Context" (DYCON)* Ref.: PID2020-115460GB-I00, (1/09/2021 - 28/02/2025), 96.800 €. Spanish Ministry of Innovation and Science. Principal Investigators: Juan Miguel Morales González and Salvador Pineda Morente.

Jun 2021 - Dec 2022 *"Mathematical Optimization Methods for Decision Making using Contextual Information" (MOMIC)* Ref.: P20-00153, (23/06/2021 - 31/12/2022), 66.000 €. Council for Economic Transformation, Industry, Knowledge and Universities (Andalusia Regional Government). Principal Investigator: Juan Miguel Morales González.

Apr 2019 - Jan 2024 *FlexAnalytics: "Advanced analytics to empower the small flexible consumers of electricity"*. Ref. 0806250180, (1/02/2018 - 31/01/2023), 1.203.125 €. European Research Council Starting Grant. Principal Investigator: Juan Miguel Morales González.

Apr 2019 - Sep 2021 *"Mathematical Methods for Data-driven Power Systems" (PowerMath)* Ref. ENE2017-83775-P, 2018-2021 (4 years), 52.030€ (+ FPI grant). Spanish Ministry of Economy, Industry and Competitiveness. Principal Investigator: Juan Miguel Morales González.

Jan 2016 - Jan 2020 *"Mathematical Optimization for Data Visualization and Decision Making"*. Ref. MTM2015-65915-R. 2016 - 2020. Spanish Ministry of Economy, Industry and Competitiveness. Principal Investigators: Rafael Blanquero Bravo and Emilio Carrizosa Priego.

Jun 2015 - Mar 2018 *"Optimización Global. Nuevos Algoritmos y Aplicaciones"* Ref. P11-FQM-7603, 2015-2018. Junta de Andalucía. Principal Investigator: Emilio Carrizosa Priego.

Jan 2015 - Dec 2015 *"Mathematical Programming: Models, Methods and Applications"*, Ref. MTM2012-36163-C06-03, 2015. Spanish Ministry of Economy, Industry and Competitiveness. Principal Investigator: Emilio Carrizosa Priego.

Organization of Conferences, Seminars and Courses

Sep 2022 Chair of session. The 3rd Spanish Young Statisticians and Operational Researchers Meeting (SYSORM).

Jul 2022 Chair of session: "Machine Learning and Mathematical Optimization: challenges and real-world applications". The 32nd European Conference on Operational Research (EURO 2022).

Jul 2021 Chair of sessions "Machine Learning and Optimization: new challenges, applications and objectives (I and II)". 31st European Conference on Operational Research (EURO 2021)

Mar 2018 Coorganizer of "Version Control Systems for Mathematical Researchers". Mathematical Research Institute of the University of Seville (IMUS). University of Seville. Seville, Spain

Dec 2015 Coorganizer of "II Pre-Doc Course on Operations Research 2015 (2nd part)". Mathematical Research Institute of the University of Seville (IMUS). University of Seville. Seville, Spain

Sep 2015 - Sep 2017 Coorganizer of the "PhD Seminar". Mathematical Research Institute of the University of Seville (IMUS). University of Seville. Seville, Spain

Jun 2015 Coorganizer of "II Pre-Doc Course on Operations Research 2015". Mathematical Research Institute of the University of Seville (IMUS). University of Seville. Seville, Spain

Recognized Academic Merits

Research 1 Recognized Six-Year Research Periods (*Sexenios*) by CNEAI/ANECA (Most recent period: 2015–2020).

Teaching	1 Recognized Five-Year Teaching Periods (*Quinquenios*) (Most recent period: 2019–2023).
Regional	1 Regional Academic Merit Tranches (*Tramos de Complementos Autonómicos*) (Last awarded: 2024).
Seniority	3 Recognized Three-Year Service Periods (*Trienios*) (Last awarded: 2024).

Membership of Societies and Groups

Since Nov 2019	Member of the research group TEP - 967 of the University of Málaga
Since Feb 2018	Member of the Spanish Society of Statistics and Operations Research (SEIO)
Jul 2014 - May 2019	Member of the research group FQM - 329 of the University of Seville

Teaching Experience

Sep 2026 - Jan 2027	**Operations Research (Investigación Operativa)**, *4th year Bachelor in Mathematics*, Sciences Faculty, University of Málaga. Subject coordinator 6 ECTS Credits Theory and Practice
Sep 2025 - Jan 2026	**Operations Research (Investigación Operativa)**, *4th year Bachelor in Mathematics*, Sciences Faculty, University of Málaga. 6 ECTS Credits Theory and Practice
Feb 2025 - Jun 2025	**Fundaments of Probability and Statistics (Fundamentos de Probabilidad y Estadística)**, *1st year Bachelor in Mathematics - 2nd year Bachelor in Mathematics and Computer Science - 2nd year Bachelor in Mathematics and Telecommunications Engineering*, Sciences Faculty, University of Málaga. Subject coordinator 6 ECTS Credits Theory and Practice
Sep 2024 - Jan 2025	**Operations Research (Investigación Operativa)**, *4th year Bachelor in Mathematics*, Sciences Faculty, University of Málaga. Subject coordinator 6 ECTS Credits Theory and Practice
Feb 2024 - Jun 2024	**Fundaments of Probability and Statistics (Fundamentos de Probabilidad y Estadística)**, *1st year Bachelor in Mathematics - 2nd year Bachelor in Mathematics and Computer Science - 2nd year Bachelor in Mathematics and Telecommunications Engineering*, Sciences Faculty, University of Málaga. Subject coordinator 6 ECTS Credits Theory and Practice

Sep 2023 - Jan 2024 **Operations Research (Investigación Operativa)**, *4th year Bachelor in Mathematics*, Sciences Faculty, University of Málaga.
Subject coordinator
6 ECTS Credits
Theory and Practice

Feb 2023 - Jun 2023 **Introduction to Probability and Statistics (Introducción a la Probabilidad y a la Estadística)**, *1st year Bachelor in Mathematics - 2nd year Bachelor in Mathematics and Computer Science - 2nd year Bachelor in Mathematics and Telecommunications Engineering*, Sciences Faculty, University of Málaga.
Subject coordinator
6 ECTS Credits
Theory and Practice

Sep 2022 - Jan 2023 **Operations Research (Investigación Operativa)**, *4th year Bachelor in Mathematics*, Sciences Faculty, University of Málaga.
Subject coordinator
6 ECTS Credits
Theory and Practice

Feb 2022 - Jun 2022 **Introduction to Probability and Statistics (Introducción a la Probabilidad y a la Estadística)**, *1st year Bachelor in Mathematics - 2nd year Bachelor in Mathematics and Computer Science - 2nd year Bachelor in Mathematics and Telecommunications Engineering*, Sciences Faculty, University of Málaga.
Subject coordinator
6 ECTS Credits
Theory and Practice

Oct 2018 - Feb 2019 **Statistics (Estadística)**, *Bachelor of Labour Relations and Human Resources (Grado en Relaciones Laborales y Recursos Humanos)*, Labour Sciences Faculty, University of Seville.
3 ECTS Credits
Theory and Practice

Oct 2018 - Feb 2019 **Statistics applied to Biology (Estadística aplicada a la Biología)**, *Bachelor of Biology*, Biology Faculty, University of Seville.
1.2 ECTS Credits
Practice

Feb 2018 - June 2018 **Introduction to Statistics (Introducción a la Estadística)**, *Bachelor of Law, Management and Public Administration (Doble Grado en Derecho y Gestión y Administración Pública)*, Law Faculty, University of Seville.
3 ECTS Credits
Theory and Practice

Feb 2018 - June 2018 **Statistics applied to Biochemistry (Estadística aplicada a la Bioquímica)**, *Bachelor of Biochemistry*, Biology Faculty, University of Seville.
0.7 ECTS Credits
Practice

Feb 2018 - June 2018 **Statistics (Estadística)**, *Bachelor of Informatics Engineering*, Informatics Faculty, University of Seville.
1.8 ECTS Credits
Practice

Feb 2017 - June 2017 **Introduction to Statistics (Introducción a la Estadística)**, *Bachelor of Law, Management and Public Administration (Doble Grado en Derecho y Gestión y Administración Pública)*, Law Faculty, University of Seville.
3 ECTS Credits
Theory and Practice

Feb 2017 - June 2017 **Statistics applied to Biochemistry (Estadística aplicada a la Bioquímica)**, *Bachelor of Biochemistry*, Biology Faculty, University of Seville.
0.7 ECTS Credits
Practice

Sep 2016 - Feb 2017 **Statistics applied to Biology (Estadística aplicada a la Biología)**, *Bachelor of Biology*, Biology Faculty, University of Seville.
2.5 ECTS Credits
Practice

Feb 2016 - June 2016 **Statistical Data Analysis (Análisis de Datos Estadśticos)**, *Bachelor of Management and Public Administration (Grado en Gestión y Administración Pública)*, Law Faculty, University of Seville.
3 ECTS Credits
Theory and Practice

Master Thesis Supervision

Jun 2026 **Kolmogorov-Arnold Geometry in Forward-Forward Networks**, *Juan Carlos Fernández Garcés*, Master in Mathematics, University of Málaga.
The Spanish version of the document is available at: `https://hdl.handle.net/10630/48090`
Calification: 10 (out of 10) with honors.

Bachelor Thesis Supervision

Sep 2026 **Kolmogorov-Arnold Geometry in Multilayer Neural Networks**, *Juan Carlos Fernández Garcés*, Bachelor in Mathematics, University of Málaga.
The Spanish version of the document is available at: `https://hdl.handle.net/10630/47200`
Calification: 9.8 (out of 10) with honors.

Nov 2025 **Optimization under Uncertainty. Stochastic and Robust Frameworks for the Knapsack Problem**, *Carolina Berrocal Enríquez*, Bachelor in Mathematics, University of Málaga.
The Spanish version of the document is available at: `https://hdl.handle.net/10630/41421`
Calification: 10 (out of 10) with honors.

Jun 2025 **SVM and Decision Trees: A Practical Application on a Real Dataset**, *Teresa Moral Fernández*, Bachelor in Mathematics, University of Málaga.
The Spanish version of the document is available at: `https://hdl.handle.net/10630/39419`
Calification: 8.5 (out of 10).

Jun 2024 **Feature Selection for Support Vector Machine**, *Antonio Luis Gómez Cuevas*, Bachelor in Mathematics, University of Málaga.
The Spanish version of the document is available at: `https://riuma.uma.es/xmlui/handle/10630/31791`
Calification: 9 (out of 10).

Sep 2023 **Valid Inequalities in Integer Programming. Application to the Knapsack Problem**, *Lourdes Albacete Maza*, Bachelor in Mathematics, University of Málaga.
The Spanish version of the document is available at: `https://riuma.uma.es/xmlui/handle/10630/27697`
Calification: 9.5 (out of 10) with honours.

Sep 2022 **Supervised Learning: Methods, Properties and Applications**, *Gema Valenzuela González*, Bachelor in Mathematics, University of Málaga.
The Spanish version of the document is available at: `https://riuma.uma.es/xmlui/handle/10630/25147`
Calification: 8.5 (out of 10).

Teaching Courses Received

Jul 2026 **Practical Introduction to AI in Teaching(Introducción Práctica a la IA en la Docencia)**, *Teacher Training and Innovation Service of the University of Málaga (Servicio de Formación e Innovación Docente de la Universidad de Málaga)*, Málaga, Spain.
1.5 hours

Nov 2022 - Jul 2024 **Novel22 (Training Program for New Teachers)**, *Teacher Training and Innovation Service of the University of Málaga (Servicio de Formación e Innovación Docente de la Universidad de Málaga)*, Málaga, Spain.
450 hours

Mar 2023 - Apr 2023 **Formative evaluation in higher education (Evaluación formativa en educación superior)**, *Teacher Training and Innovation Service of the University of Málaga (Servicio de Formación e Innovación Docente de la Universidad de Málaga)*, Málaga, Spain.
20 hours

Nov 2021 **New Virtual Campus Tool for Plagiarism Detection (Nueva Herramienta de Campus Virtual para la Detección de Plagio)**, *Teacher Training and Innovation Service of the University of Málaga (Servicio de Formación e Innovación Docente de la Universidad de Málaga)*, Online format.
3 hours

Nov 2020 **Microsoft Teams for Teaching (Microsoft Teams para la Docencia)**, *Teacher Training and Innovation Service of the University of Málaga (Servicio de Formación e Innovación Docente de la Universidad de Málaga)*, Online format.
6 hours

Mar 2020 **Scientific Research and Methodology, beyond the Master thesis (Investigación y Metodología Científica, Más Allá del TFM)**, *Higher Institute of Psychological Studies (Instituto Superior de Estudios Psicológicos - ISEP)*, Rafael Bisquerra, Barcelona, Spain (Online format).
3 hours

Scientific Dissemination Activities

Mar 2025 **Educathon 25 - Girls in Science**, *International Day of Women and Girls in Science*, University of Málaga, Málaga, Spain.
Invited researcher

Apr 2024 **Night in the classrooms (La noche en las aulas)**, *Algorithmic adventure: Exploring coding processes with the help of Asunción*, Secondary school "Isaac Albéniz" and "Maristas", Málaga, Spain.
Invited researcher

Apr 2024 **Recording a podcast**, *Workshop on scientific-engineering professions to make female referents more visible*, Primary school "Flor de Azahar", Cártama, Málaga, Spain.
Talk

Feb 2024 **Educathon 24 - Girls in Science**, *International Day of Women and Girls in Science*, University of Málaga, Málaga, Spain.
Invited researcher

Feb 2023 **Educathon 23 - Girls in Science**, *International Day of Women and Girls in Science*, University of Málaga, Málaga, Spain.
Invited researcher

Feb 2019 **An Introduction to Mathematical Reasoning with Applications to Induction and Contradiction Proofs**, *Karumi*, *M. Asunción Jiménez Cordero*, Madrid, Spain.
Talk

Nov 2018 **Science Week 2018**, *Mathematical Research Institute of the University of Seville (IMUS). University of Seville.*, Seville, Spain.
Participant

Other Training Courses

May 2026 **Guide to Applying for Erasmus Mundus Joint Master Degrees (Guía para Solicitar Másteres Conjuntos Erasmus Mundus)**, *University of Málaga*, Online, 3 hours.

Jun 2025 **Emotional Intelligence: The Art of Managing Emotions (Inteligencia Emocional: el Arte de Saber gestionar las Emociones)**, *University of Málaga*, Online, 1.5 hours.

Apr 2025 **The Risks of Digitalization (Los Riesgos de la Digitalización)**, *University of Málaga*, Online, 1.5 hours.

Mar 2025 **UMA Virtual Campus Tools (Herramientas del Campus Virtual de la UMA)**, *University of Málaga*, Online, 20 hours.

Feb 2025 **Coordination of Curricular Internships via the Ícaro Platform (Coordinación de las Prácticas Curriculares a través de la Herramienta Ícaro)**, *University of Málaga*, Online, 3 hours.

Feb 2025 **PROA Program for Faculty and Departments (PROA para Docentes y Departamentos)**, *University of Málaga*, Online, 5 hours.

Feb 2025 **Prevention of Psychosocial Risks and Mental Health in the Working Population (Prevención de Riesgos Psicosociales y Salud Mental de la Población Trabajadora)**, *University of Málaga*, Online, 1.55 hours.

Nov 2024 **Workshop on Academic Accreditation and Six-Year Research Evaluations: Sciences (Taller sobre Acreditación y Sexenios: Ciencias)**, *University of Málaga*, Online, 2 hours.

Nov 2024 **Psychological First Aid (Primera Ayuda Psicológica)**, *University of Málaga*, Online, 1 hour.

Jun 2024 **Accreditation Workshop: Science (Taller sobre acreditación: ciencias)**, *University of Málaga*, Online, 2 hours.

Apr 2024 **New law on universities, regulatory challenges in the promotion of internationalization (Nueva ley de universidades, retos regulatorios en la promoción de la internacionalización)**, *University of Málaga*, Online, 10 hours.

Apr 2024 **Stress management techniques. Breathing and mindfulness (Técnicas de gestión del estrés. Respiración y mindfulness)**, *University of Málaga*, Online, 1.5 hours.

Jan 2024 **Effective communication for conflict management (Comunicación eficaz para la gestión de conflictos)**, *University of Málaga*, Online, 1.5 hours.

Oct 2023 **Workshop on APC (Article Process Charges) management (Taller sobre Gestión de APC (Article Process Charges))**, *University of Málaga*, Online, 1.5 hours.

Oct 2023 **Vancouver Standard - Workshop for the Elaboration of Citations and Bibliographical References (Norma Vancouver - Taller para la Elaboración de Citas y Referencias Bibliográficas)**, *University of Málaga*, Online, 1.5 hours.

Apr 2023 **Formative Evaluation in Higher Education (Evaluación Formativa en Educación Superior)**, *University of Málaga*, Málaga, Spain, 20 hours.

Apr 2023 **Gender Equality as a University Principle: Towards the I+ Label (La Igualdad de Género como Principio Universitario: hacia el Distintivo I+)**, *University of Málaga*, Málaga, Spain, 15 hours.

Nov 2022 **APA Standard - Workshop for the Elaboration of Citations and Bibliographical References (Norma APA - Taller para la Elaboración de Citas y Referencias Bibliográficas)**, *University of Málaga*, Online, 1.5 hours.

Oct 2022 **Workshop on Visibility of Scientific Production (Taller sobre Visibilidad de la Producción Ciéntifica)**, *University of Málaga*, Online, 1.5 hours.

Oct 2022 **Workshop on Open Research Data (Taller sobre Datos Abiertos de Investigación)**, *University of Málaga*, Online, 1.5 hours.

Feb 2022 **Workshop on Creating a Research Data Management Plan (Taller sobre Creación de un Plan de Gestión de Datos de Investigación)**, *University of Málaga*, Online, 2.5 hours.

Abr 2021 **Open Science Global Mini Symposium**, *Imfahe Foundation and the Doctoral school of the University of Valladolid*, Online.

April 2020 **Citation Booster: Enhacing the reach, readership and visibility of your publication**, *Charlesworth Author Services*, Online.

Other Merits

Sep 2017 Collaborator of the Apache Spark project "Roma. A sentiment analysis tool based on SVM". Available at `https://github.com/pedrovgs/Roma`, `https://pedrovgs.github.io/Roma/` and `http://blog.karumi.com/a-story-of-tweets-and-sentiments/`

Languages

Spanish Native

English B2 level. Certificate by Official Languages School of Seville (Escuela Oficial de Idiomas de Sevilla). June 2010

French B2 level. Certificate by Official Languages School of Seville (Escuela Oficial de Idiomas de Sevilla). June 2012

German B2 level. Certificate by Official Languages School of Seville (Escuela Oficial de Idiomas de Sevilla). June 2017

Computational Skills

- Cinderella
- Fortran
- Git
- Github
- Haskell
- LaTeX
- Mathematica
- Matlab
- Python
- R
- Sage
- Scilab
- SPSS
- Xpress

Research Interests

Mixed integer nonlinear programming, Inverse problems, Combinatorial problems, Model selection, Feature selection, Classification, Regression, Functional Data Analysis, Bilevel Optimization, Artificial Intelligence